

Final report for grant A49, design of the Big Quiet Air Purifier

Overview:

This project went well as far as it was taken, however we terminated it early to roll the funding that remained into the Big Quiet Fan mass production. This is because it became apparent that the taped together air purifiers actually work quite well, and the other aspects that differentiate a BQAP from a taped together assembly (which I dub an Extra Quiet CR box or EQ-CR box) are not really that important, but at the same time are costly.

However, the anemometer and the internet of things control box were a good success. Secondly, I did learn a lot which I do plan to eventually hopefully apply to build additional prototypes, and if the market is amenable, to sell alongside the big quiet fan, however it doesn't look like a low hanging fruit.

One finished air purifier is now in operation in a classroom at the University of Waterloo. A second one is at Cite Des Hospitaliers, a coworking space, in Montreal. A third is at Foulab, a hacker space in Montreal. A fourth was located at a social central called Lachoppe, but they foolishly threw it in the trash, an absurd thing to do, illustrating that the machine alone will never be enough, anti-propaganda is important as well.

The school board of Connecticut has apparently funded taped together CR boxes for all its 40,000 classrooms, at about 11.5 million USD or something, so apparently taped together boxes are adequate even in that sensitive context. Thus, focussing on the fan seems like a good first step.

I developed solutions for the physical structure, holding filters of various sizes with the same physical structure, and getting a large number of filters integrated into the machine to reduce pressure drop. I developed the 6 filter internal V box, which nobody else seems to have thought of before but works well, and is in general the winning configuration. Even a good solution for the handles at the corners is done.

I hope to use extruded aluminum in the future as the wood was of marginal strength and appearance.

I realized that because of how the details stack up, adding more filters generally decreases the cost per cubic meter. I realized cost per cubic meter is a good way to think about things.

I developed a good approach to a cosmetic cover, and have ideas to improve it further. Solutions for tamper proofing were also found (durable yet low cost grilles and also mounting it high up). I have ideas for putting the fan in pushing mode to further increase CADR and mounting it near the ceiling to leverage displacement flow.

Concepts for embedding the whole system in a false ceiling are interesting.

Budget:

Total budget was \$30,000. Not quite all of this was expended. This may indicate that it would be a tight squeeze to make 30 units. However the plan was supposed to be to actually sell the units so that part would have been mostly self funding. I think I struck a pretty good balance.

			CAD	USD
labor cost			25060	18159.42029
sum of all costs with labor			31072	22515.94203

Estimated impact

The designs are published, as well as some information about the general lessons learned. The filter configuration with the diagonal internal V and 6 filters is applicable to taped together boxes, allowing increased performance while also reducing cost.

Once we have a good supply of the critical ingredient, the fans, it will be time to pick this up again, or perhaps someone else will. There are a number of examples of entrepreneurs making kits for air purifiers, and this both builds upon and enables that further. I have offered Northbox a sample control box and anemometer assembly so he can make his air purifiers remote controllable and monitorable (including the filter clogage levels).

Many users will not be happy with taped together CR boxes, so this is a good start on a good option with the same net performance, albeit at a higher upfront cost, and with the same great low running costs. Filter changes are easier and the finished assembly may be possible to get UL listing for, encouraging uptake in hospitals and other tightly regulated spaces.